

TrES-3b

R-0133

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230525 · created 2026-07-16T06:29:15+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460089.8737 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.021
Transit depth (Rp/Rs)^2	3.86 +/- 0.81 [%]
Orbital Inclination (inc)	81.4 +/- 0.45
Airmass coefficient 1 (a1)	0.8943 +/- 0.0089
Airmass coefficient 2 (a2)	0.106 +/- 0.063
Scatter in the residuals of the lightcurve fit is	1.0 %
Best Comparison Star	#3 - [219, 131]
Optimal Aperture	4.39
Optimal Annulus	13.45
Transit Duration (day)	0.0544 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.021 vs 0.16309 (deviation 1.36σ)
Duration fit vs published	0.0544 d vs 0.0567 d (deviation 0.37σ)
Mid-time O–C	-0.0 min
Depth SNR	8.1
Residual scatter	1.0 %

Light curve

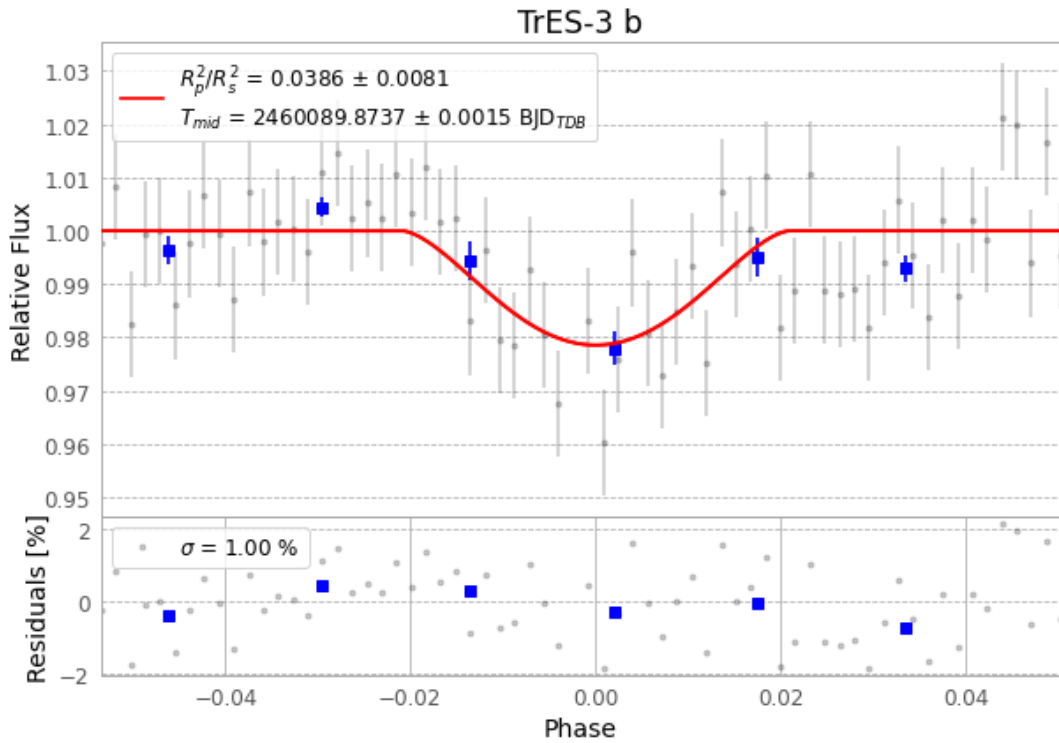


Figure 1 — FinalLightCurve_TrES-3 b_2023-05-25.png (SHA-256 a566ec546ef8d33e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [292, 291], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e48ff2bde961f99e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

66 FITS frames (43 MB), TRES-3230525071215.FITS → TRES-3230525102715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230525.log (0540720e4db9...), FinalLightCurve_TrES-3 b_2023-05-25.png (a566ec546ef8...), FinalLightCurve_TrES-3 b_2023-05-25.csv (fd8ba7638029...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 06:29Z.